Highlights

ATLAS: Anytime-Valid Monitoring and Certification of Neural World Models

Quang-Vinh Dang

- Anytime-valid monitoring of deployed neural world models, resolved by horizon
- Separate guarantees for false alarms (frontier) and for safety (certified horizon)
- Valid on policy-steered streams; raw-error monitors inflate false alarms there
- A drift gate cuts planning cost under actuator faults for a neural dynamics model
- Tolerance calibration decides real-video false alarms; negative results reported

ATLAS: Anytime-Valid Monitoring and Certification of Neural World Models

Quang-Vinh Dang[a,*]

[a]*British University Vietnam, Hung Yen, Vietnam*

Abstract

A world model supports planning only while its imagined rollouts stay close to what the environment would do, and that closeness depends on the rollout horizon and changes during deployment. Offline scores are computed once and follow neither effect. We study sequential monitoring and certification of deployed neural world models with error guarantees that hold uniformly over time and need no assumption on how the planner generates the data. ATLAS runs a betting test martingale for each horizon against the null that the horizon is ε-faithful, giving a *revocation frontier* that controls false alarms, and a confidence sequence on the faithfulness gap, giving a *certified horizon* that controls the opposite, safety-relevant error. We prove time-uniform validity, a correct confidence sequence for the running average of conditional means with a per-horizon total-variation certificate, and a finite-sample detection-delay bound. On a simulator with known truth, a first-order delay law predicts measured delay within 20% and the certified bound is never violated. On a closed-loop stream steered by a model-based planner, CUSUM and a conformal martingale on raw prediction error alarm in every run while ATLAS stays at or below 0.013. Gating a planner built on a neural dynamics model with ATLAS lowers the mean post-fault cost over three faults from 34.4 to 8.2–13.9 with no cost in fault-free operation, though it reacts more slowly than CUSUM. On real video, a frozen-encoder model reaches zero false alarms and 0.9–1.0 detection at a tolerance margin of one standard deviation, while a small convolutional autoencoder does not. Two negative results: pooling overlapping horizon chains does not recover the data they discard, and a tolerance calibrated on pooled data voids the conditional guarantee.

Keywords: world models, anytime-valid inference, e-processes, sequential testing, model-based reinforcement learning, neural network monitoring, change detection

1. Introduction

A model-based agent imagines several futures, dozens of steps deep, at every decision, and acts on the one that looks best. That is sensible while the world model matches the environment. Then a joint wears, friction changes, a payload shifts, or the agent's own improving policy carries it into states the model never saw. The imagined futures stay confident, nothing in the monitoring stack fires, because the only number anyone computed was an offline score on held-out data before deployment, and the failure is found by its consequences.

*Corresponding author. ORCID: 0000-0002-3877-8024

Email address: vinh.dq4@buv.edu.vn (Quang-Vinh Dang)

Two properties make the faithfulness of a world model awkward to certify. It *decays with the horizon*: a rollout applies the model recursively, so a small one-step error compounds, and a model that is nearly perfect one step ahead can be useless twenty steps ahead. Faithfulness is therefore a profile over horizons rather than a scalar. It also *drifts over deployment time*, through non-stationarity, the sim-to-real gap, and the feedback loop of model-based reinforcement learning, where the policy generating the data is optimized through the model under test. The data stream is then *endogenous*: it depends on the object being audited.

Current practice scores world models with Fréchet Video Distance (Unterthiner et al., 2018), reconstruction error, or return gaps, all computed once on a fixed set. Conformal prediction (Vovk et al., 2005) and its online variants (Gibbs and Candès, 2021, 2024) are the usual tools for distribution-free uncertainty, but they target marginal coverage of a single prediction. Classical change detectors (CUSUM, Page–Hinkley, windowed tests) control an average run length, so their false-alarm probability grows with monitoring time. What is wanted instead is a running statement, "at round t, rollouts of this model are trustworthy up to horizon h at tolerance ε," whose error probability is controlled uniformly over all rounds, resolved by horizon, indifferent to whether the model predicts pixels or embeddings, and sound when the planner steers the data.

Approach.. We build this from the testing-by-betting machinery of anytime-valid inference (Shafer, 2021; Ramdas et al., 2023; Waudby-Smith and Ramdas, 2024). At each round the model emits, for each horizon, an action-conditioned predictive distribution; reality later reveals the outcome. A strictly proper score (Gneiting and Raftery, 2007) turns the pair into a bounded statistic. For each horizon a test martingale bets against the null that the horizon is ε-faithful, which is a *conditional-mean* null, one constraint given the whole past; that is why no exchangeability or independence is needed and why the construction survives policy endogeneity. The same betting machinery, run as a confidence sequence, bounds the faithfulness gap from above. These two outputs answer different questions and control different errors:

- the *revocation frontier* $h^\star(t)$, the largest horizon whose e-processes have not yet crossed $1/\alpha$, is a monitor: a faithful horizon is revoked with probability at most α over all time (an error of availability);
- the *certified horizon* $\hat{h}_\tau(t)$, the largest horizon whose confidence-sequence bound on total variation is below a budget τ, is a certificate: with probability at least $1-\alpha$, every horizon the planner is allowed to use has window-average discrepancy at most τ (an error of safety).

A planner that clips its imagination depth to $\min(h^\star, \hat{h}_\tau)$ inherits both guarantees, jointly with probability at least $1-2\alpha$ (the certificate is an average over a window of logged rounds, see Theorem 2). The two must not be conflated: a bound on false revocations says nothing about trusting an unfaithful horizon.

What is and is not new.. Betting e-processes and confidence sequences are not new, and neither is sequential forecast evaluation with e-values (Section 1.1). We do not claim a new inference principle. The contributions are the problem formalization for deployed neural world models, the analysis of what breaks when the target is multi-step and the planner steers the data, and evidence on when the construction works and when it does not:

1. **Formalization and multi-step validity** (Sections 2–3). A horizon-indexed, conditional-mean faithfulness null with a tolerance, a statistic whose conditional mean is the object of interest, and an analysis of why consecutive h-step residuals cannot be bet on directly. We validate on non-overlapping chains, show that pooling offset chains does not help, and give a recipe for neural world models whose multi-step predictive is available only by sampling: the planner commits to an action block, which makes the predictive well defined for every horizon inside the block (Section 3.4).
2. **Two dual guarantees** (Section 4): time-uniform false-revocation control; a confidence sequence for the running average of *conditional* means (the constant-mean construction does not certify it); a windowed variant; and a per-horizon total-variation certificate with a simulation-lemma-style value-gap consequence.
3. **A finite-sample delay bound** replacing the asymptotic statement, with a horizon-resolved growth rate that predicts measured delay across horizons.
4. **Evidence** (Section 5): ground-truth validation; a closed-loop study where the planner steers the stream; a controlled study in which a neural-dynamics planner is *gated* by the frontier under three faults, which measures whether the certificate improves control; replicated real-video experiments with three world-model classes and competitor monitors on identical streams; and ablations that include the cases where design choices fail.

1.1. Related work

Anytime-valid inference and betting. Ville's inequality (Ville, 1939), game-theoretic probability and e-values (Shafer and Vovk, 2019; Vovk and Wang, 2021; Ramdas et al., 2023), safe testing (Grünwald et al., 2024), and betting confidence sequences (Waudby-Smith and Ramdas, 2024; Howard et al., 2021) supply our tools. Betting has been applied to two-sample testing (Shekhar and Ramdas, 2024), to auditing fairness of a deployed classifier (Chugg et al., 2023), and to tracking the risk of a deployed model under shift (Podkopaev and Ramdas, 2022).

Sequential forecast evaluation. The closest prior work evaluates probabilistic forecasts sequentially with e-values: calibration tests for probability forecasts (Arnold et al., 2023), comparison of forecasters (Henzi and Ziegel, 2022b; Choe and Ramdas, 2024). These establish that time-uniform inference on forecast scores is possible, and one of them has direct bearing on our setting: a correction to Henzi and Ziegel (2022b) withdraws a supermartingale property for horizons above one, the same overlap phenomenon that we analyze in Section 3.3. They address calibration or relative performance of a forecaster at a fixed lead time. We instead test an absolute tolerance null at every horizon of an action-conditioned simulator, certify the gap from above, and study streams in which the actions depend on the model.

Conformal prediction and change detection. Conformal test martingales (Vovk et al., 2003) monitor exchangeability, and online conformal methods (Gibbs and Candès, 2021, 2024; Tibshirani et al., 2019) relax it for coverage. Change detectors based on cumulative sums, Page–Hinkley, windowing, and Bayesian run-length posteriors (Page, 1954; Bifet and Gavaldà, 2007; Raab et al., 2020; Adams and MacKay, 2007) are tuned to an average run length, and the e-detector framework (Shin et al., 2024) makes the false-alarm guarantee nonparametric. We compare against all of these in Section 5.6.

World models and the simulation lemma. World models range from latent-imagination agents (Ha and Schmidhuber, 2018; Hafner et al., 2023) and joint-embedding predictive

architectures (Assran et al., 2023; Bardes et al., 2024; Assran et al., 2025) to interactive and physical-AI simulators (Bruce et al., 2024; NVIDIA et al., 2025). Their evaluation is offline. The simulation lemma (Kearns and Singh, 2002; Janner et al., 2019; Lobel et al., 2024) turns one-step model error into a value gap, with a known one-step error. We replace the known error by a time-uniform certified bound.

2. Problem formulation

2.1. Deployment protocol

A world model M is deployed online with a planner π. At round t the agent is in a (possibly latent) state x_t and the planner chooses actions $a_{t:t+H-1}$ up to a maximum horizon H, using M for imagined rollouts. For each horizon $h \le H$ the model produces an action-conditioned predictive distribution

$$P^M_{t,h} := P^M\big(\cdot \mid x_t,\, a_{t:t+h-1}\big) \in \Delta(\mathcal{Y}) \tag{1}$$

over a target space $\mathcal{Y}$ (observations, states, or a learned embedding). Reality reveals the h-step outcome $Y^{(h)}_t$ at time $t{+}h$. Let $\mathcal{F}_{k-1}$ be the deployment information available at the issue time s_k of the k-th prediction of the horizon under consideration: states, committed actions, planner internals, and all previous outcomes (on a non-overlapping chain, Section 3.3, previous predictions are resolved before the next is issued). We assume nothing about independence or exchangeability of the stream. A variable is *predictable* if it is $\mathcal{F}_{k-1}$-measurable.

2.2. Scores, divergence, and the faithfulness null

Fix a strictly proper, negatively oriented score S (Gneiting and Raftery, 2007) and write $d_S(Q,P) = \mathbb{E}_{Y\sim Q}[S(P,Y)] - \mathbb{E}_{Y\sim Q}[S(Q,Y)] \ge 0$ for its divergence, which vanishes iff $P = Q$. For the log score $d_S = \mathrm{KL}$ and Pinsker gives $\mathrm{TV} \le \sqrt{d_S/2}$; for the energy score d_S is a squared energy distance; for a Gaussian predictive the standardized squared error $(y-\mu)^\top \Sigma^{-1}(y-\mu)$ is χ^2_d under calibration, which lets ATLAS score a predicted *embedding* against the realized embedding without decoding to pixels. We assume a modulus: there is a nondecreasing concave ϕ and a probability metric ρ with $\rho(Q,P) \le \phi(d_S(Q,P))$.

Let $Q_{s,h} = \mathrm{Law}(Y^{(h)}_s \mid \mathcal{F}_s)$ be the true conditional law.

Definition 1 (ε-faithfulness)**.** Horizon h is ε_h-faithful if $d_S(Q_{s,h}, P^M_{s,h}) \le \varepsilon_h$ for every issue time s.

We operationalize it through a bounded per-round statistic $Z_k \in [z_{\mathrm{lo}}, z_{\mathrm{hi}}]$ for each horizon, issued at s_k and resolved at $s_k{+}h$, whose conditional mean $\mu_k := \mathbb{E}[Z_k \mid \mathcal{F}_{k-1}]$ is tied to the divergence in two directions.

Definition 2 (Calibrated and certifiable statistics)**.** The statistic is *calibrated at tolerance* ε_h if $d_S(Q_{s_k,h}, P^M_{s_k,h}) \le \varepsilon_h$ implies $\mu_k \le \varepsilon_h$ (used for the *test*). It is *certifiable with constants* $(c, \bar\rho)$ if $d_S(Q_{s_k,h}, P^M_{s_k,h}) \le c\,\mu_k + \bar\rho$ almost surely (used for the *certificate*). Certifiability is a statement about the statistic whose conditional mean is bounded; if the computed statistic is clipped at z_{hi}, the clipping lowers the mean and $\bar\rho$ must absorb the difference (negligible in our simulator, where $z_{\mathrm{hi}} = 6$ and the neglected tail has probability about 10^{-4}).

Example 1 (Reference score excess)**.** With a predictable reference R_k and $Z_k = S(P^M_{s_k,h}, Y) - S(R_k, Y)$ we have $\mu_k = d_S(Q, P^M) - d_S(Q, R_k)$. It is calibrated at every ε_h since $d_S(Q, R_k) \geq 0$, and certifiable with $c = 1$ and $\bar{\rho} \geq \sup d_S(Q, R_k)$, the suboptimality of the reference. A poor reference makes the test conservative and the certificate loose.

Example 2 (Standardized Gaussian excess)**.** For a Gaussian predictive $\mathcal{N}(\mu_M, \Sigma_M)$ take $Z_k = \frac{1}{d}(y - \mu_M)^\top \Sigma_M^{-1}(y - \mu_M) - 1$, clipped to $[-1, z_{\text{hi}}]$. Without clipping, if the true conditional law has the same covariance and differs in mean by m, then $\mu_k = \frac{1}{d} m^\top \Sigma_M^{-1} m$ and $d_S = \text{KL} = \frac{d}{2}\mu_k$, so $c = d/2$, $\bar{\rho} = 0$, and $\mu_k = 0$ exactly under calibration. The conditional mean is *pivotal*: it does not depend on the state or the action. If the covariance is mismatched, only the one-directional calibration property is retained.

2.3. The endogeneity obstruction

The outcomes $\{Y_t^{(h)}\}$ are neither independent nor exchangeable, and the planner is optimized through M, so the state distribution that the monitor sees is a function of the model. Two things follow. A reference or threshold calibrated offline, on data the planner did not generate, describes a different distribution from the one met in deployment (Section 5.3 measures how much that matters). And a fixed constant tolerance is only meaningful if the statistic's conditional mean does not itself depend on where the planner goes. A conditional-mean null constrains one expectation given the entire past, so for a horizon-one statistic it holds for any predictable planner and betting rule, with no model of how the planner behaves. For $h > 1$ the action sequence $a_{s:s+h-1}$ on which the predictive is conditioned must be committed at issue time (open-loop or a pre-committed plan); if later actions react to intermediate states, the realized-action predictive is not the conditional law and the null can fail for a faithful model. Our endogeneity experiments use $h = 1$. What it does not do is guarantee power: a planner may drift to states where the same error is hard to detect, which we quantify in Section 5.3.

3. The ATLAS method

3.1. E-processes and Ville's inequality

An *e-variable* for a null is a nonnegative random variable with expectation at most one under it. A *test supermartingale* is a process $W_0 = 1$, $W_k = \prod_{j \leq k} e_j$ with $\mathbb{E}[e_j \mid \mathcal{F}_{j-1}] \leq 1$ under the null. The engine of time-uniform validity is classical.

Lemma 1 (Ville 1939)**.** *If $(W_k)_{k \geq 0}$ is a nonnegative supermartingale with $W_0 = 1$ then $\mathbb{P}(\sup_k W_k \geq 1/\alpha) \leq \alpha$.*

3.2. The per-horizon betting e-process

Fix a horizon h and a statistic $Z_k \in [z_{\text{lo}}, z_{\text{hi}}]$ calibrated at ε_h (Definition 2), so that the faithfulness null implies

$$\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] \leq \varepsilon_h \qquad \text{for all } k. \tag{2}$$

For a predictable bet $\lambda_k \in [0, \lambda_{\max}]$ with $\lambda_{\max} = 1/(z_{\text{hi}} - z_{\text{lo}})$ set

$$e_k := 1 + \lambda_k(Z_k - \varepsilon_h), \qquad W_K^{(h)} := \prod_{k \leq K} e_k. \tag{3}$$

Proposition 1 (Per-round validity)**.** *If (2) holds and $\varepsilon_h \in [z_{\text{lo}}, z_{\text{hi}}]$ then $e_k \geq 0$ and $\mathbb{E}[e_k \mid \mathcal{F}_{k-1}] \leq 1$; hence $W^{(h)}$ is a test supermartingale.*

Proof. $e_k \geq 1 - \lambda_{\max}(\varepsilon_h - z_{\text{lo}}) \geq 0$, and $\mathbb{E}[e_k \mid \mathcal{F}_{k-1}] = 1 + \lambda_k(\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] - \varepsilon_h) \leq 1$ because λ_k is predictable. □

The bet governs power, not validity. We use a uniform mixture over a grid $\lambda^{(1)}, \ldots, \lambda^{(m)} \in (0, \lambda_{\max}]$: a convex combination of test supermartingales is a test supermartingale, so $W_K^{(h)} = \frac{1}{m} \sum_j \prod_{k \leq K}(1 + \lambda^{(j)}(Z_k - \varepsilon_h))$ is valid, parameter-free, and adapts to an unknown margin at a cost of $\log m$ in the log-wealth. Section 5.7 compares it with a fixed bet and a plug-in bet.

3.3. Multi-step validity: why chains, and why they cannot be pooled

If a horizon-h prediction is formed at *every* round, consecutive residuals share the innovations in a window of length $h - 1$, so the residual stream is moving-average of order $h - 1$. Conditioning on the past reveals part of the current residual and (2) fails in general even for a faithful model. The same effect underlies the overlap problem in sequential multi-step forecast evaluation (Henzi and Ziegel, 2022a). We restore (2) by running each horizon on *non-overlapping chains* of issue times $c, c+h, c+2h, \ldots$: the k-th chain element is resolved before the next is issued, so $\mathcal{F}_{k-1}$ contains everything the next residual could depend on. The cost is real: horizon h receives a factor h fewer updates.

One might try to win these updates back by running all h offset chains and rejecting when any crosses its threshold (a union bound, costing $\log h$). That is valid but it does not pool evidence: each chain sees only a $1/h$ share of the data and its wealth grows at the same rate per update as the single chain, so taking the first of h crossings buys at most a small order-statistic gain that the $\log h$ in the threshold cancels. Averaging the chains' wealth would pool, but chains live on different filtrations, so the average is an e-value at fixed times and not an e-process (an optional-stopping argument does not go through). Section 5.2 measures the interleaved union: it is valid and no faster than the single chain. We therefore keep the simple estimator and treat the factor h as the intrinsic price of certifying long horizons directly.

3.4. Neural world models: sampled predictives and committed action blocks

A neural world model rarely has a closed-form multi-step predictive. We take it by sampling: the stochastic model (for instance a Gaussian head on the next latent) is rolled forward n times under the action sequence, and the standardized excess of Example 2 is computed from the sample mean and covariance on its non-degenerate subspace. Two requirements for validity follow from Section 2. First, the action sequence must be fixed at issue time. A planner that replans at every step reacts to intermediate noise, so the realized actions are not independent of the outcome and the null can fail even for a faithful model. We therefore let the planner *commit* to an open-loop block of length L and replan only at block boundaries; for each horizon $h \leq L$ the block yields $\lfloor L/h \rfloor$ non-overlapping chain elements, issued at block offsets $0, h, 2h, \ldots$, each conditioned on actions that were known when it was issued. Second, the statistic's conditional mean is tied to the divergence only if the sampled predictive's covariance is adequate, which is why the tolerance is calibrated on held-out nominal closed-loop episodes of the same controller (Section 5.4).

The commit length is the natural place for the gate: the planner runs with $L = \min(L_{\max}, h^\star(t))$, checks the frontier after every executed step, and aborts the remaining block and enters a safe fallback when $h^\star \leq 1$ (Algorithm 1).

Algorithm 1 ATLAS: monitoring and certification of a deployed world model

input: horizons $\{1, \dots, H\}$, tolerances $\{\varepsilon_h\}$, level α, bet grid $\{\lambda^{(j)}\}$, budget τ, constants $(c, \bar{\rho})$
initialize wealth $W^{(h)} \leftarrow 1$, `revoked`$(h) \leftarrow$ false, and a confidence sequence $\mathrm{CS}^{(h)}$ for each h
for each round $t = 1, 2, \dots$ **do**
  for each horizon h whose chain element resolves at t **do**
    form the bounded statistic Z; update $W^{(h)} \leftarrow \sum_j \frac{1}{m} \Pi_j (1 + \lambda^{(j)}(Z - \varepsilon_h))$ and $\mathrm{CS}^{(h)}$
    if $W^{(h)} \geq H/\alpha$ **then**
      `revoked`$(h) \leftarrow$ true
    end if
  end for
  $h^\star(t) \leftarrow$ largest prefix with no revoked horizon; $\hat{h}_\tau(t) \leftarrow$ largest prefix with $\widehat{\mathrm{TV}}_i \leq \tau$
  output the gate $\min(h^\star(t), \hat{h}_\tau(t))$
end for

3.5. Two outputs: the revocation frontier and the certified horizon

Revocation frontier (monitor).. Run one e-process per horizon at level α/H and revoke horizon h at $\tau_h = \inf\{k : W_k^{(h)} \geq H/\alpha\}$, permanently (Ville concerns the event of *ever* crossing). Define

$$h^\star(t) := \max\{h : \text{ no horizon } h' \leq h \text{ is revoked by round } t\}, \tag{4}$$

with $h^\star = 0$ if horizon one is revoked. The prefix rule encodes that a broken short horizon casts doubt on longer ones, and matches the use as a depth clip. The guarantee (Theorem 1) is *no false revocation*: a faithful horizon is lost with probability at most α.

Certified horizon (certificate).. In estimation mode the same machinery gives a time-uniform upper bound $U_t^{(h)}$ on the running average of conditional means of horizon h (Lemma 2), hence, for a certifiable statistic, an upper bound on the average divergence and, through the modulus, on total variation: $\widehat{\mathrm{TV}}_h(t) := \min\{1, \phi(cU_t^{(h)} + \bar{\rho})\}$. For a budget τ the certified horizon is

$$\hat{h}_\tau(t) := \max\{h : \ \widehat{\mathrm{TV}}_i(t) \leq \tau \text{ for all } i \leq h\}. \tag{5}$$

The guarantee (Theorem 2) is the opposite error: with probability at least $1 - \alpha$, simultaneously over rounds and horizons, the true average discrepancy of every horizon inside $\hat{h}_\tau(t)$ is at most τ. The planner's gate is $\min(h^\star(t), \hat{h}_\tau(t))$. The two do not substitute for each other: $h^\star$ can be high on a model that is mediocre but stable, and $\hat{h}_\tau$ can be low on a faithful model that has not yet accumulated enough evidence.

Change detection.. To detect a change in faithfulness rather than test a fixed null we use the Shiryaev–Roberts e-detector (Shin et al., 2024), $R_t = (1 + R_{t-1})e_t$ (with the mixture bet), which accumulates evidence over all candidate changepoints so that wealth lost before the change cannot delay the alarm. Algorithm 1 summarizes the procedure.

4. Theory

Throughout, statistics are bounded, $Z_k \in [z_{\mathrm{lo}}, z_{\mathrm{hi}}]$ with width $w := z_{\mathrm{hi}} - z_{\mathrm{lo}}$, and bets and references are predictable. No independence or exchangeability is assumed.

4.1. No false revocation

Theorem 1 (Uniform validity)**.** *Fix $\alpha \in (0,1)$. If horizon h satisfies* (2)*, then $\mathbb{P}(\exists K : W_K^{(h)} \geq 1/\alpha) \leq \alpha$, uniformly over all predictable planners and betting rules. Running the H horizons at level α/H each gives $\mathbb{P}(\exists t,\ \exists h \in \mathcal{H}_0 :\ h$ revoked by round $t) \leq \alpha$ for the set $\mathcal{H}_0$ of faithful horizons.*

Proof. By Proposition 1 and linearity, $W^{(h)}$ is a nonnegative supermartingale with $W_0^{(h)} = 1$ under the null, using only (2) and predictability. Ville's inequality bounds $\mathbb{P}(\sup_K W_K^{(h)} \geq H/\alpha) \leq \alpha/H$ for each horizon, and a union bound over $|\mathcal{H}_0| \leq H$ horizons gives the simultaneous statement. □

The guarantee is conservative: under the null the wealth is a supermartingale that tends to drift down, so empirical rates sit far below α (Section 5.1). It controls false *revocations*; it says nothing about trusting an unfaithful horizon, which is the job of the certificate below.

4.2. A confidence sequence for the average conditional mean

The simulation-lemma-style bound needs the running average $m_K = \frac{1}{K}\sum_{k\leq K} \mu_k$ of the *conditional* means, which vary over the deployment. A betting confidence sequence for a *constant* mean does not certify it: its proof tests $\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] \geq u$ for every k, which is not implied by an average bound. We use an empirical-Bernstein supermartingale instead.

Lemma 2 (Confidence sequence for the average conditional mean)**.** *Let $\widehat{Z}_k \in [z_{\mathrm{lo}}, z_{\mathrm{hi}}]$ be predictable, $\psi(\lambda) = -\log(1-\lambda) - \lambda$, $V_K = \sum_{k\leq K} \big((Z_k - \widehat{Z}_k)/w\big)^2$, and fix a grid $\lambda_1, \ldots, \lambda_n \in (0,1)$ with prior weights ν_j. With probability at least $1-\alpha$, simultaneously for all $K \geq 1$ and all j,*

$$m_K \ \leq\ U_K := \bar{Z}_K + \frac{w}{\lambda_j K}\Big(\psi(\lambda_j)V_K + \log\frac{1}{\alpha\nu_j}\Big), \tag{6}$$

so the minimum over j is a valid upper confidence sequence. The width is of the empirical-Bernstein order $\sqrt{\hat{\sigma}_K^2 \log(1/\alpha)/K} + w\log(1/\alpha)/K$.

Proof. Rescale to $[0,1]$. For $y \geq -1$ and $\lambda \in [0,1)$, $\exp(\lambda y - \psi(\lambda)y^2) \leq 1 + \lambda y$ (Fan et al., 2015). Put $y = \widehat{x}_k - x_k \in [-1,1]$. Then $\mathbb{E}[\exp(\lambda(\mu_k - x_k) - \psi(\lambda)(x_k - \widehat{x}_k)^2) \mid \mathcal{F}_{k-1}] \leq e^{\lambda(\mu_k - \widehat{x}_k)}(1 + \lambda(\widehat{x}_k - \mu_k)) \leq 1$, for *any* value of the conditional mean μ_k. So $M_K(\lambda) = \exp(\lambda\sum_k(\mu_k - x_k) - \psi(\lambda)V_K)$ is a nonnegative supermartingale, and so is $\sum_j \nu_j M_K(\lambda_j)$. Since $\nu_j M_K(\lambda_j)$ is dominated by the mixture, Ville's inequality gives $\mathbb{P}(\exists K, j :\ \nu_j M_K(\lambda_j) \geq 1/\alpha) \leq \alpha$; rearranging gives (6). □

Corollary 1 (Windows)**.** *Restart the sequence at block starts $s_i = ib$ and give start s_i the level $\alpha/((i+1)(i+2))$. Then with probability at least $1-\alpha$, simultaneously for all starts and all end times, the average conditional mean over $(s_i, t]$ is at most the corresponding bound. Reading the window with the latest start $s_i \leq t - w_0$ gives an anytime-valid bound on the* recent *average over a window of length in $[w_0, w_0 + b)$ (for $t < w_0$ it is the whole history).*

Proof. Apply Lemma 2 at each start with level $\alpha_i = \alpha/((i+1)(i+2))$ and use $\sum_i \alpha_i \leq \alpha$. □

The windowed form matters: the full-history average moves slowly after a change, so a certificate meant to follow the present must be read over a window; the price is the larger $\log(1/\alpha_i) \approx 2\log i$ term.

4.3. The certified horizon and a simulation-lemma consequence

Theorem 2 (Anytime-valid certified horizon)**.** *Suppose each horizon's statistic is certifiable with constants $(c, \bar{\rho})$ (Definition 2) and $\rho = \mathrm{TV}$ with modulus ϕ. Apply Lemma 2 (or Corollary 1) to the chain of each horizon $h \le H$ at level α/H, giving bounds $U_t^{(h)}$ for the average over a chain index set $\mathcal{S}_{h,t}$ (full history or a window). Then with probability at least $1-\alpha$, simultaneously for all rounds t and horizons $h \le H$,*

$$\frac{1}{|\mathcal{S}_{h,t}|} \sum_{k \in \mathcal{S}_{h,t}} \mathrm{TV}\big(Q_{s_k,h}, P^M_{s_k,h}\big) \;\le\; \min\big\{1,\; \phi\big(cU_t^{(h)} + \bar{\rho}\big)\big\}. \tag{7}$$

Consequently, with probability at least $1-\alpha$, for all t and all $h \le \hat{h}_\tau(t)$ the left side is at most τ. The statement is about the window average *over logged issue times: a short spike inside a long window can be diluted below τ, and it concerns the logged action sequences, not a counterfactual imagined policy.*

Proof. By certifiability the average divergence over $\mathcal{S}_{h,t}$ is at most $c\,m + \bar{\rho}$ with m the average conditional mean, which Lemma 2 bounds by $U_t^{(h)}$ for all t, h with probability $1-\alpha$ after a union bound over H horizons. By the modulus, concavity of ϕ (Jensen) and monotonicity, the average of $\mathrm{TV} \le \phi(d_S)$ is at most $\phi(c\,m + \bar{\rho}) \le \phi(cU_t^{(h)} + \bar{\rho})$; total variation is at most one. □

Corollary 2 (Value gap)**.** *Let rewards $r \in [0, R_{\max}]$ be a function of the outcome and let actions be held fixed (open-loop), as in the action-conditioned predictive. For discount weights $w_i \ge 0$, on the event of Theorem 2 the weighted sum of window-averaged per-step reward gaps satisfies $\sum_{i \le h} w_i\, G_i(t) \le R_{\max} \sum_{i \le h} w_i \min\{1, \phi(cU_t^{(i)} + \bar{\rho})\}$, where $G_i(t)$ is the average of $|\mathbb{E}_M r(Y^{(i)}) - \mathbb{E}_Q r(Y^{(i)})|$ over horizon i's chain index set.*

Proof. For fixed actions $|\mathbb{E}_P r - \mathbb{E}_Q r| \le R_{\max}\mathrm{TV}(P, Q)$; apply (7) for each i. □

Remark 1 (Relation to the classical simulation lemma)**.** The classical lemma (Kearns and Singh, 2002) converts a *known* one-step error into an h-step value gap, paying a compounding factor such as $\gamma/(1-\gamma)^2$. Here each horizon's own chain is certified, so there is no propagation step and no concentrability coefficient (the price is that the certified quantity is the discrepancy on the logged action sequences, not on counterfactual rollouts of a new policy); the price is that horizon h is certified from a $1/h$ share of the rounds. When the gap profile is constant across issue times in the window, $\sum_i w_i G_i$ is the average discounted return gap. We are not aware of a time-uniform version of this statement in the literature.

4.4. Detection delay

Take the e-detector R_t built from the mixture bet. For the alarm time $T = \inf\{t : R_t \ge 1/\alpha\}$ the classical argument gives the false-alarm guarantee, and a martingale concentration argument gives a finite-sample delay.

Theorem 3 (Finite-sample detection delay)**.** *Let $X_k = Z_k - \varepsilon$ (so $|X_k| \le w$, the width of the statistic's range).* (a) *Under the null, $\mathbb{E}_\infty[T] \ge 1/\alpha$.* (b) *Suppose that after a changepoint τ, $\mathbb{E}[X_k \mid \mathcal{F}_{k-1}] \ge \Delta > 0$ and $\mathbb{E}[X_k^2 \mid \mathcal{F}_{k-1}] \le v$ for all $k > \tau$, and that the grid is $\lambda_j = j/(mw)$, $j = 1, \ldots, m$ (the one the implementation uses) and $\Delta \ge 2v/(mw)$. Let $\lambda^\star = \min\{\Delta/(2v),\, 1/(2w)\}$. For every $\delta \in (0,1)$ and*

$$n \;\ge\; N := \max\Big\{ \frac{8\log(m/\alpha)}{\lambda^\star \Delta},\; \frac{(80v + 8w\Delta)\log(1/\delta)}{\Delta^2} \Big\}, \tag{8}$$

we have, for every history, $\mathbb{P}(T-\tau > n \mid \mathcal{F}_\tau, T > \tau) \le \delta$. *Note* $8/(\lambda^\star\Delta) \le 16(v/\Delta^2+w/\Delta)$. (c) *For horizon* h *on a non-overlapping chain, with per-chain margin* Δ_h *and second moment* v_h*, the delay in rounds is at most* $h(N_h+1)$.

The proof is in Appendix Appendix A. When the first term of (8) dominates and the variance term is the larger one ($v \gg w\Delta$), $N \approx 16\,v\log(m/\alpha)/\Delta^2$, which is within a constant factor of the Lorden–Lai rate $2\sigma^2\log(1/\alpha)/\Delta^2$ for a Gaussian-like pair with the same mean shift and variance (Lorden, 1971; Lai, 1998) (the comparison is of a $1-\delta$ quantile with an expected delay, so it is a statement about order, not about constants; the second term of (8) dominates for $\delta \lesssim 0.2$). The bound is loose in practice: on the simulator of Section 5.2 it gives N in the thousands for horizon one against a measured delay near 500. When the bet saturates ($v \lesssim w\Delta$) the delay scales as w/Δ instead of $1/\Delta^2$. Part (c) is the horizon-resolved law: horizon h pays a factor h for fewer updates and gains through a larger margin Δ_h as the shift compounds along the rollout. Section 5.2 tests the first-order prediction $h(\log(H/\alpha)+\log m)/D_h^\star$ with $D_h^\star = \sup_\lambda \mathbb{E}\log(1+\lambda Z)$.

Remark 2 (The e-process alone)**.** For the plain e-process W started at 1, the same argument holds with $\log(m/\alpha)$ replaced by $\log(m/(\alpha W_\tau))$, where W_τ is the wealth at the changepoint: wealth lost before the change delays the alarm. The Shiryaev–Roberts form removes this term because it contains the term started at $\tau{+}1$ at full wealth.

5. Experiments

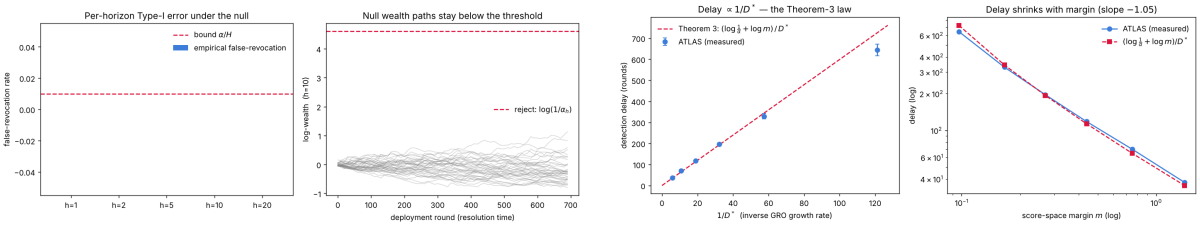
The study has five parts. A simulator with known ground truth checks each guarantee against the truth (Section 5.1), including the horizon-resolved delay law (Section 5.2). A closed-loop stream in which a model-based planner steers the state tests the endogeneity claim (Section 5.3). A controlled study with a neural dynamics model asks whether gating the planner with ATLAS improves control (Section 5.4). Replicated real-video experiments with three world-model classes compare ATLAS with standard monitors on identical streams (Sections 5.5–5.6). Ablations, including the cases where design choices fail, close the section (Section 5.7). Unless stated, $\alpha = 0.05$, the bet is a mixture over 20 values, and horizons are run on non-overlapping chains at level α/H. Code and per-experiment JSON outputs are in the repository.

5.1. Ground-truth validation

The environment is a stable linear-Gaussian latent system whose world model uses the pre-change parameters, so it is exactly faithful until an unmodelled force of size f is added at a known round (Appendix Appendix B). The statistic is the standardized Gaussian excess of Example 2.

No false revocation.. Over 400 null deployments of 700 rounds, no horizon among $\{1,2,5,10,20\}$ was ever revoked, against a per-horizon bound of 0.01 and an any-horizon bound of 0.05. In the horizon-resolved study below (seven horizons, 400 null runs of 2,400 rounds) the any-horizon rate is 0.005. The rates sit far below the bound, as expected for an e-process that is a supermartingale under the null.

Detection delay.. Sweeping the drift margin over 100 seeds, the measured delay matches $(\log(1/\alpha)+\log m)/D^\star$ to within 13% at the smallest margin and 8% elsewhere, with delay $\cdot\, D^\star \approx 5.2$, and scales as $1/D^\star$ (Figure 1). In this range the bet saturates, $D^\star \propto$ margin, and delay scales as one over the margin; the $1/\text{margin}^2$ regime of Theorem 3 requires margins below about 0.07.



(a) Per-horizon false revocation under the null (left) and null wealth paths (right).

(b) Delay against the predicted $(\log \frac{1}{\alpha} + \log m)/D^\star$ (left) and against $1/D^\star$ (right).

Figure 1: Ground-truth validation of Theorems 1 and 3 on the simulator.

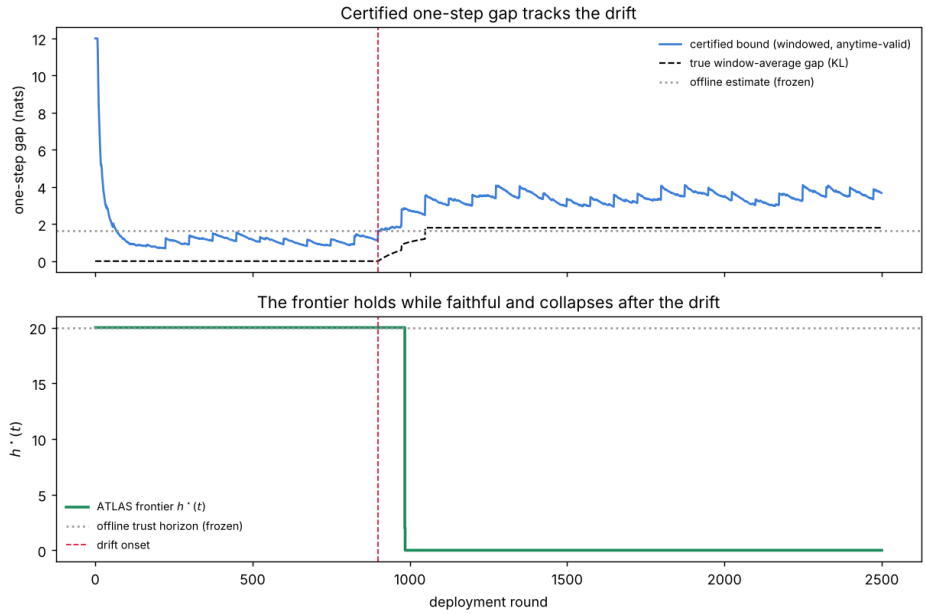


Figure 2: A dynamics drift opens at round 900. Top: the windowed, anytime-valid bound on the one-step gap (solid) against the true window-average gap (dashed) and a frozen offline estimate (dotted). Bottom: the revocation frontier $h^\star(t)$.

The frontier and the certified gap.. With a drift of size 0.6 opening at round 900 (Figure 2), horizons $1, 2, 3, 5, 8, 12, 20$ are revoked $85, 120, 84, 105, 132, 144, 220$ rounds after the change; the frontier falls from 20 to 0, first dropping 84 rounds after the change, while an offline estimate frozen at deployment stays at 20. The windowed certified gap (window 150) is never below the true window-average gap in this one deployment (an illustration, not a coverage test; coverage is in Table 1). It is also loose: 1.63 nats before the change when the true gap is zero and 3.67 after against a true 1.80. A certificate is not a point estimate; it costs data (Section 6).

The certified horizon.. Using one chain per horizon with windows of 400 rounds, the certified total-variation bound of Theorem 2 was checked against the closed-form truth over 150 deployments, at checkpoints before the drift and 300, 800 and about 2,000 rounds after it, for $h \in \{1, 3, 8\}$ (Table 1). It was never violated (0/150 at every cell; the guarantee allows 5%) and its slack, bound over truth, is between 1.6 and 3.0 after the change. The slack comes mostly from Pinsker's inequality, which is loose for a mean shift; each cell uses a confidence sequence at level $\alpha/(2H)$ and the check is at four fixed times, not a supremum over t.

Table 1: Certified total-variation bound against the exact value, 150 deployments, drift size 0.25 opening at round 500. Violations: 0 in every cell. Median slack = bound / truth.

	true TV_h	slack +300	slack +800	slack + $\sim$ 2000
$h = 1$	0.307	2.83	2.17	2.29
$h = 3$	0.456	2.96	2.19	2.19
$h = 8$	0.635	2.13	1.58	1.58

Table 2: Rounds from the drift to revocation, by horizon. Prediction uses only the post-change statistic law. The interleaved estimator takes the first crossing over all h offset chains at the threshold raised by $\log h$.

horizon h	1	2	3	5	8	12	20
growth rate $D_h^\star$	0.016	0.031	0.044	0.067	0.100	0.136	0.186
predicted delay	502	521	543	594	635	699	852
measured, single chain	507	484	497	523	537	585	732
measured, interleaved	507	476	494	554	617	712	923

5.2. Horizon-resolved delay and the pooling negative result

Theorem 3(c) predicts that horizon h pays a factor h for fewer updates and gains through a larger margin as the drift compounds. We measure the rounds from the change to revocation for each horizon (150 runs, drift 0.25) and compare with $h(\log(H/\alpha)+\log m)/D_h^\star$, where $D_h^\star = \sup_\lambda \mathbb{E}\log(1+\lambda Z)$ is estimated by Monte Carlo from the post-change stationary statistic (Table 2, Figure 3).

The first-order prediction is within 20% of the measurement at every horizon and errs on the conservative side; it is a heuristic law (growth rate of the post-change statistic), not the finite-sample bound N of Theorem 3, which is 9–15$\times$ loose here. Two features are worth stating plainly. First, in this system $D_h^\star$ grows roughly in proportion to h, so the factor-h loss is almost exactly offset and delay in rounds is nearly flat in h (between 484 and 732); long horizons are not slower *here*, but nothing in the theory says that must hold for other dynamics. Second, the interleaved estimator, which was meant to win back the data discarded by non-overlap, is no faster and is slower at long horizons (923 against 732 rounds at $h = 20$), as the pooling argument of Section 3.3 predicts. Under the null both estimators stay valid (any-horizon false revocation 0.005 and 0.003).

5.3. Policy-endogenous streams

We test the claim that exchangeability- and calibration-based monitors lose validity when the planner steers the state, on a closed-loop latent world model with state-dependent noise in which the world model is exactly faithful, so every alarm under the null is a false alarm (Appendix Appendix B). Four planners generate the stream: exogenous exploration (*exo*); a certainty-equivalent controller run through the model toward a low-noise state (*low*) or toward a high-noise state (*high*); and a monitor-aware planner (*adv*) that runs ATLAS's own e-process on the stream it produces and moves to the high-noise region whenever the wealth is positive. Under the null the wealth is usually below one, so this planner sits mostly in the low-noise region (its mean noise level equals that of *low*); it is a weak adversary and the point of the experiment is the monitors on raw error, not a worst case. The mean noise level induced is 0.24, 0.15, 0.33 and 0.15 respectively. Each monitor sees either the *raw* squared prediction error, the quantity practitioners monitor, or the

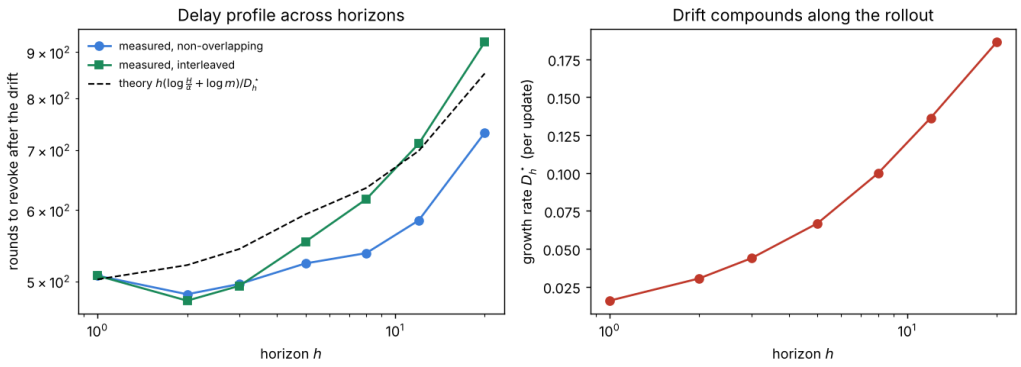


Figure 3: Left: measured delay (single chain and interleaved) against the prediction. Right: the growth rate $D_h^\star$ rises with the horizon because the drift accumulates along the rollout.

Table 3: False-alarm rate on a closed-loop stream with a faithful model: at 400 rounds / by 1,200 rounds, 300 runs per cell. Tunable detectors are calibrated on exogenous data to 5% at 400 rounds. "raw" monitors see squared prediction error, "std" monitors the standardized statistic.

stream	monitor	exo	low	high	adv
raw	CUSUM	.07/.22	.00/.00	**1.00/1.00**	.40/.49
raw	Page–Hinkley	.04/.38	.00/.00	.01/.13	.45/.73
raw	ADWIN	.12/.53	.00/.04	.00/.01	.54/.73
raw	KSWIN	.06/.18	.00/.06	.00/.01	.20/.40
raw	Conformal MG	.09/.10	.00/.00	**1.00/1.00**	.44/.49
std	CUSUM	.09/.28	.07/.23	.09/.25	.08/.26
std	Page–Hinkley	.08/.56	.08/.54	.06/.56	.07/.52
std	ADWIN	.03/.19	.05/.20	.05/.16	.04/.17
std	KSWIN	.02/.08	.03/.08	.03/.06	.02/.08
std	Conformal MG	.03/.03	.03/.03	.01/.01	.02/.02
std	**ATLAS**	**.00/.01**	**.00/.01**	**.00/.01**	**.00/.01**

model-*standardized* statistic that ATLAS uses. Tunable detectors are calibrated to a 5% false-alarm rate at 400 rounds on a separate exogenous calibration deployment, the usual offline protocol, then run for 1,200 rounds under each planner (300 runs each).

Three patterns appear in Table 3. (i) Calibrated offline on exogenous data, raw-error monitors do not carry over: CUSUM and the conformal martingale alarm in every run when the planner heads for the high-noise region, and under the monitor-aware planner every raw monitor reaches 0.40–0.73; under the low-noise planner the same monitors alarm almost never, which would mask a real drift. (ii) On the standardized stream the exchangeability-based conformal martingale is valid for *every* planner. That is not an accident: when the model is exactly faithful the standardized statistic is pivotal, identically distributed whatever the planner does, so the planner cannot hurt a monitor that sees it. The change detectors on that stream still inflate with monitoring time (Page–Hinkley $0.08 \to 0.56$). The honest summary is that conformal monitors are fine on a pivotal statistic and ATLAS's advantage is that it needs no pivotality and no tuning, only a conditional-mean bound. (iii) ATLAS never exceeded 0.013 under any planner. Because the standardized statistic is pivotal, this cannot fail for any planner; we have not tested a non-pivotal statistic under steering, and ablation D (exogenous) shows that case can fail.

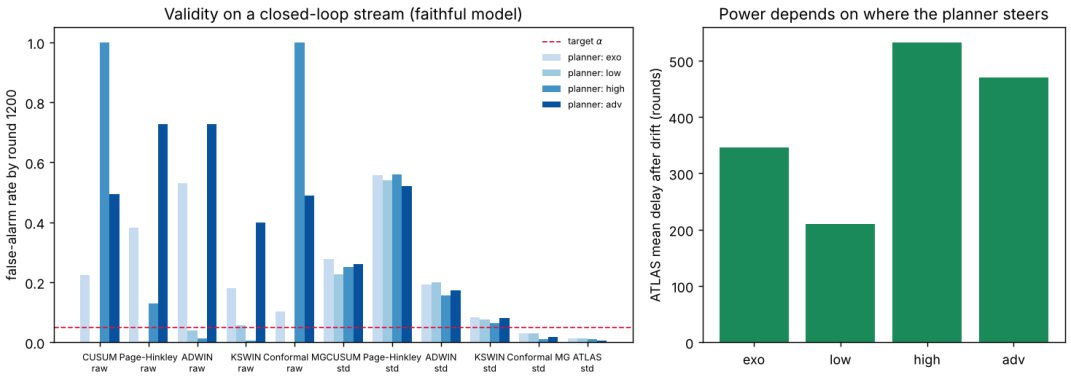


Figure 4: Left: false-alarm rate by round 1,200 for each monitor and planner on the closed-loop stream. Right: ATLAS's mean detection delay after a drift under each planner.

Power depends on the planner.. After an unmodelled drift of size 0.4 at round 400, ATLAS detected the drift in every run with mean delay 346, 211, 532 and 471 rounds under *exo*, *low*, *high* and *adv*. The planner that seeks high-noise states shields the drift by roughly 1.5× against exogenous exploration, since the same mean shift moves the standardized statistic less where the noise is larger. The conformal martingale on the standardized stream is slower by 1.6–2.2× in every setting and misses the drift in 3–5% of runs. Validity under endogeneity is a theorem; power under endogeneity is an empirical quantity that this study measures for one family of planners and does not bound in general.

5.4. Gated planning with a neural world model

The previous experiments certify a model; this one asks whether the certificate improves control. A damped point mass with quadratic drag must track a target on a circle. A heteroskedastic MLP (next velocity as a Gaussian, positions by known kinematics) is trained on random-action transitions of the nominal system only, and a gradient-based planner through its mean dynamics commits to action blocks of length L. Long horizons matter in this task: the nominal tracking cost falls from 0.75 at a committed horizon of 4 to 0.15 at 16. Halfway through each 400-step episode a fault changes the true dynamics while the model keeps its nominal behavior: the actuator gain drops to 0.4 (*gain*), the x-action changes sign (*flip*), or a constant force pushes along x (*wind*). Controllers share the model and the planner: fixed horizons 4 and 16 without a gate; the ATLAS gate of Section 3.4 with horizons $\{1, 2, 4, 8, 16\}$ at tolerance margins $k \in \{0.25, 0.5, 1, 2\}$ (tolerances from 12 separate nominal episodes per model; $k = 1$ was the default fixed in advance from the real-video protocol); a CUSUM gate on the one-step standardized residual, tuned to a 5% false-alarm rate on the same calibration episodes; an oracle gate that brakes at the true fault time; and brake-always. The safe fallback is zero action. There are five trained models and 20 episodes per cell.

Four observations from Table 4. (i) No ATLAS gate ever fired in a fault-free episode, for any margin, and nominal cost is unchanged; the CUSUM gate fired in 5% of fault-free episodes. (ii) The flip fault makes the ungated long-horizon planner diverge (cost 99.2). The ATLAS gate reduces it to 37.8 at $k = 1$, 21.1 at $k = 0.5$ and 13.3 at $k = 0.25$ (mean alarm delay 70, 44 and 29 steps; 130 and cost 73.8 at $k = 2$), which is a 62–87% reduction for $k \leq 1$ but well short of the CUSUM gate (4.7, delay 4 steps) and the oracle (5.5). ATLAS is slower than CUSUM here because its bounded bets need several informative updates before wealth crosses the threshold. (iii) For the *gain* and *wind* faults, braking is

Table 4: Gated planning with a neural world model. Mean stage cost (lower is better), 20 episodes per cell, bootstrap 95% intervals in brackets. "No fault" is the mean cost over the second half of a fault-free episode and "false trigger" the fraction of fault-free episodes in which a gate fired. Gate controllers switch to the safe fallback (zero action) when they alarm. Cost before the fault is 0.154 for every long-horizon controller.

	no fault	false trigger	gain ×0.4	x-axis flipped	wind	mean (3 faults)
fixed horizon 4	0.98	—	1.73 [1.7,1.8]	64.35 [59.3,69.3]	4.88 [4.2,5.6]	23.7
fixed horizon 16 (ungated)	0.16	—	1.06 [1.0,1.1]	99.21 [96.0,102.5]	2.80 [2.5,3.1]	34.4
ATLAS gate, $k = 0.25$	0.16	0.00	1.40 [1.2,1.6]	13.34 [11.8,15.1]	9.74 [5.3,15.0]	8.2
ATLAS gate, $k = 0.5$	0.16	0.00	1.13 [1.0,1.2]	21.09 [18.8,23.8]	4.05 [3.0,5.3]	8.8
ATLAS gate, $k = 1$	0.16	0.00	1.06 [1.0,1.1]	37.81 [33.8,42.3]	2.87 [2.6,3.2]	13.9
ATLAS gate, $k = 2$	0.16	0.00	1.06 [1.0,1.1]	73.84 [68.0,80.7]	2.80 [2.5,3.1]	25.9
CUSUM gate	0.21	0.05	3.14 [2.6,3.7]	4.67 [3.9,5.5]	64.65 [58.9,69.7]	24.2
oracle gate	0.16	—	3.48 [2.9,4.1]	5.47 [4.6,6.4]	66.58 [63.7,69.5]	25.2
brake always	4.37	—	4.37 [3.3,5.6]	4.37 [3.3,5.6]	92.81 [86.4,99.6]	33.9

the wrong response: the ungated planner stays near cost 1.06 and 2.80, but braking costs 4.4 and 92.8. CUSUM and the oracle gate brake on any change and are far worse on wind (64.6 and 66.6). The ATLAS gate at $k \geq 1$ does not fire for these faults (0 of 20 episodes), at $k = 0.5$ it fires in 10–20% of them, and at $k = 0.25$ in 80–90%, paying for it (gain 1.40, wind 9.74). (iv) Averaged over the three faults, the ATLAS gate has the lowest cost for $k \leq 1$ (8.2, 8.8, 13.9) against 34.4 ungated, 24.2 for CUSUM and 25.2 for the oracle.

The reading of this table must be careful. The ranking depends on the fault suite and on the fallback: zero action is right for a mis-wired actuator and wrong under a persistent disturbance, and no fault-agnostic fallback exists in this system. The margin k trades responsiveness against needless fallbacks, and it is a statistical tolerance on the model's predictive accuracy rather than a bound on task cost, so a gate can ignore a fault the task would have wanted noticed and react to one it would have tolerated. What the experiment does show is narrower: with no per-fault tuning, a tolerance-based anytime-valid gate can avoid the catastrophic outcome of an unfaithful long-horizon model at no nominal cost, whereas a gate that reacts to any change pays for false and unnecessary triggers.

5.5. Real video: replicated protocol

Three real shifts are studied with a PCA-plus-linear-dynamics latent world model: Moving MNIST (Srivastava et al., 2015) native speed to double speed, KTH (Schuldt et al., 2004) walking to running, and KTH walking outdoors to indoors. Each of 30 repetitions draws a fresh random split into training, calibration, in-distribution deployment and shifted deployment clips, refits the world model, and sets the tolerance $\varepsilon_h = \bar{z}_h + k\, s_h$ from the calibration clips (mean $\bar{z}_h$ and standard deviation s_h of the clipped statistic, with $k \in \{0, 0.25, 0.5, 1\}$). Competing detectors are tuned on a separate set of 30 repetitions to a 5% false-alarm rate over the in-distribution deployment length L. We report false alarms over L clips and over $3L$ (in-distribution clips resampled with replacement, standing in for longer monitoring), and detection rate and delay in clips after the shift, with bootstrap 95% intervals. False-alarm rates over L and $3L$ clips come from independent resamples, so the L rate can exceed the $3L$ rate by chance. The tolerance is estimated from the same population, so Theorem 1 does not apply to these runs, and no confidence sequence or certificate is run on real data.

The picture is mixed, and we report it as such. With the tolerance set at the calibration mean ($k = 0$), ATLAS's false-alarm rate over $3L$ clips is 0.43 on Moving MNIST, 0.47 on KTH walking→running, and 0.50 on KTH outdoor→indoor, all above α: pooled-mean

calibration does not satisfy the conditional null (Section 5.7, ablation D). Adding a margin repairs this at a cost in power. On Moving MNIST $k \geq 0.25$ gives no false alarm in 30 repetitions and detection in every repetition, with a delay of 4.6–6.6 clips. On KTH, subject-to-subject heterogeneity makes the statistic far less stable and calibration clips are few: $k = 0.5$ still gives 0.13 false alarms over $3L$ clips for walking→running, and $k = 1$ brings it to 0.03 while detection drops to 0.70; on the subtler outdoor→indoor shift $k = 0.5$ gives 0.03 false alarms and detects in 0.77 of repetitions. The false-alarm rates for $k \geq 0.5$–1 are close to α but nothing here is a guarantee, because the tolerance is estimated. The competing monitors fare worse on false alarms on these real streams: the conformal martingale alarms in 0.43–0.70 of in-distribution repetitions over L clips (exchangeability fails across the temporal and subject structure of the stream), CUSUM in 0.97 on KTH walking→running, and ADWIN and KSWIN, tuned to 5% on separate repetitions, either miss the shift entirely (Moving MNIST, KTH walking→running) or detect it in 5–6 clips (domain shift). Tuning a sensitivity to a target false-alarm rate on a limited number of streams is itself unstable, which is the practical case for a method whose false-alarm control does not rely on it. We read the table as evidence that the construction is usable on real video when the tolerance is set with a margin and validated, not as evidence that false-alarm control transfers automatically.

World-model classes.. Table 6 repeats the protocol with two neural world models (10 repetitions each): a convolutional autoencoder with a learned latent dynamics network, and a JEPA-style model with a frozen ImageNet ResNet-18 encoder, PCA-reduced, plus a learned latent dynamics network. The predictive covariance of both is re-estimated on held-out calibration clips. The result depends strongly on the model. The frozen-encoder model is quiet and fast at $k = 1$ (zero false alarms over $3L$ clips on both KTH shifts, detection 1.00 and 0.90, delay 3.0 and 8.3 clips), better than the linear model on the same shifts at the same margin (detection 0.70 and 0.33). The small convolutional autoencoder is not usable on KTH walking→running: it overfits the training subjects, its in-distribution statistic is non-stationary across subjects, and false alarms stay at 0.40 over $3L$ clips for every margin while detection falls from 0.70 to 0.40 (on the domain shift its false alarms are 0.10–0.20 but it detects only 0.30–0.80). On Moving MNIST it is conservative (no false alarms with $k \geq 0.25$ but delay near 50 clips, against 5 for the linear model). A certificate is only as good as the representation under it; ATLAS does not repair a weak world model, it reports it. Competing monitors on the same streams are not better: the conformal martingale alarms in 0.50–1.00 of $3L$ in-distribution streams and CUSUM in 0.20–0.60.

5.6. Comparison with competitor monitors on the simulator

On the ground-truth simulator we also compare ATLAS with CUSUM and Page–Hinkley (Page, 1954), ADWIN (Bifet and Gavaldà, 2007), KSWIN (Raab et al., 2020), Bayesian online changepoint detection (Adams and MacKay, 2007), the e-detector (Shin et al., 2024), and a conformal test martingale (Vovk et al., 2003), all on the same one-step statistic stream, with tunable detectors calibrated to 5% at 400 rounds (Table 7). The anytime-valid tests keep false alarms below α at three times the calibration horizon, while every tuned detector inflates (Page–Hinkley from 0.05 to 0.53); the tuned detectors' shorter delay comes from alarming more readily. The conformal martingale is also valid but slower (77.9 against 53.8 rounds) and, as Section 5.3 shows, loses validity when fed non-pivotal scores on a steered stream.

5.7. Ablations and negative results

Table 8 collects five ablations at horizon one (400 null and 200 alternative runs per cell, independent samples for each row, so identical configurations such as "mixture" and "step" differ by Monte Carlo noise; drift 0.25 at round 500 unless noted).

The mixture is close to the best fixed bet without knowing it (A): a bet that is too small is slow and a large fixed bet is slightly faster but is the one that depends on guessing the margin. Tolerance (B) behaves as designed: when ε is at or above the true gap the revocation rate is at most α; when ε is below it the model really is outside tolerance and revocation grows, slowly, because the gap is tiny. A ramp (C) is detected late, and a short transient is mostly missed: a cumulative e-process is not a detector of brief excursions, and users who care about transients should use a windowed variant.

Ablation D is the most important caution. The log-score advantage over a climatology reference has a *state-dependent* conditional mean, because the reference ignores the state. Setting ε to its pooled null mean makes the null hold on average but not conditionally, and validity fails (0.253 false revocations against a 0.05 guarantee). A constant ε that does hold conditionally (a conservative bound) is so large that the drift is never detected. The standardized Gaussian excess has a pivotal conditional mean and avoids the dilemma. The same issue applies to real video: the tolerances calibrated on pooled calibration clips in Section 5.5 are only empirically safe, which is why the margin k matters there. With heavy-tailed noise (E) the standardized statistic stays valid, since only the covariance enters its conditional mean, and detection is slower.

6. Discussion

What the evidence supports.. Time-uniform false-revocation control held in every setting we tried, including a planner that watches the monitor and a stream whose state distribution is a function of the model. A first-order delay law predicts measured delay across horizons to within 20% on a system where the post-change statistic is known (the finite-sample bound of Theorem 3 is 9–15× loose there). The certified total-variation bound was never violated and was off by a factor of 1.6 to 3 after a change. On real video the construction detects every shift with a zero margin, at false-alarm rates of 0.43–0.50 for the linear model; margins that bring false alarms to about 0.03 cost detection on the linear model (0.33–0.70), but a frozen-encoder neural model reaches zero false alarms with detection 0.90–1.00 at the same margin, and a weak convolutional autoencoder cannot be repaired by any margin. With a neural dynamics model in a closed loop, a tolerance-based gate removed most of the catastrophic cost of an unfaithful long-horizon model at no fault-free cost, while a gate that reacted to any change did worse on average. The tolerance and the quality of the representation decide the trade-off.

Limitations..

- *The tolerance is a modeling choice and the dangerous one.* The guarantee needs $\mathbb{E}[Z_k \mid \mathcal{F}_{k-1}] \le \varepsilon_h$ conditionally. A constant tolerance set to the pooled in-distribution mean of a statistic whose conditional mean varies with the state does not satisfy it (ablation D: 0.253 false revocations; real video with $k = 0$: Section 5.5). The standardized Gaussian excess is pivotal and has no such problem, but only for models that output a calibrated predictive covariance. On real video we calibrate ε_h from held-out clips with a margin; that is empirically safe in our replications and not a guarantee.

- *Certificates are expensive.* Pinsker's inequality squares twice on the way from the mean of the statistic to total variation, so certifying a budget τ needs on the order of τ^{-4} rounds at one horizon and h times more at horizon h. In the simulator, windows of a few hundred rounds certify a gap of the size of our drifts but not a small budget on a faithful model (certified 1.63 nats against a true 0 in Figure 2). The certified horizon is therefore informative after large changes and conservative otherwise.
- *Long horizons are intrinsically costly.* Non-overlapping chains give horizon h a $1/h$ share of the rounds; we found no valid way to pool the discarded rounds (Section 3.3). In our simulator a growing margin offsets this; in general it need not.
- *Power is not guaranteed under steering.* A planner heading for states where the same error is harder to see delays detection by about $1.5\times$ in our study; we do not bound this in general.
- *Transients and ramps.* A cumulative e-process detects sustained changes; a 150-round pulse was detected in 17% of runs and a ramp is detected late.
- *Scope of the real-video evidence.* The replicated real-video study uses three world-model classes on three shifts (Moving MNIST speed, KTH gait and domain) with 10–30 repetitions; the shifts are large benchmark shifts rather than subtle drifts. Earlier single runs on KITTI driving and UZH-FPV drone video are not replicated and are not used in any claim; subtle drifts and larger models remain untested.
- *The certificate is an average over logged rounds.* It can be diluted by a short spike inside its window, and it concerns the logged action sequences rather than a new policy's imagined rollouts. No certificate is run on real data here.
- *Multi-step claims need committed actions.* For $h > 1$ the validity argument needs the action sequence fixed at issue time; the endogeneity experiments use $h = 1$ and a pivotal statistic.
- *The gate evidence is one system.* Section 5.4 gates a planner on a point mass with three faults, five trained models and 20 episodes per cell. The result depends on the fault suite and on the fallback (zero action is wrong under a persistent disturbance), the gate is slower than a CUSUM gate when a fault is abrupt, and the tolerance is a statistical bound on predictive accuracy, not on task cost. The certified horizon $\hat{h}_\tau$ was not used in the gate because it is too conservative at the data sizes in an episode.

Practical guidance.. Use a statistic whose conditional mean is pivotal if the model provides a calibrated predictive covariance; otherwise add a margin to ε_h and validate it by replaying held-out in-distribution streams for longer than the deployment horizon. Read the certified horizon over a window rather than the full history. Use the revocation frontier as the alarm and the certified horizon as the gate; do not treat the first as evidence for the second.

Future work.. Valid pooling of overlapping horizon chains; a tolerance derived from a downstream control-cost budget rather than predictive accuracy; fault-aware fallbacks; windowed e-detectors for transients; and replicated studies of larger neural world models on driving and aerial video.

Reproducibility.. Code, seeds, and the JSON outputs of every table are in the public repository. No new data were collected; the real-video experiments use the public Moving MNIST and KTH datasets.

Appendix A. Proof of Theorem 3

(a) Average run length.. Each component detector $R_t^{(j)} = (1 + R_{t-1}^{(j)})e_t^{(j)}$ satisfies $\mathbb{E}[R_t^{(j)} \mid \mathcal{F}_{t-1}] \le 1 + R_{t-1}^{(j)}$ under the null, so the mixture $R_t = \sum_j \nu_j R_t^{(j)}$ does too, and $R_t - t$ is a supermartingale. Optional stopping at $T \wedge n$ gives $\mathbb{E}[R_{T\wedge n}] \le \mathbb{E}[T \wedge n]$. Since $R_{T\wedge n} \ge \frac{1}{\alpha}\mathbf{1}\{T \le n\}$, $\mathbb{P}(T \le n)/\alpha \le \mathbb{E}[T]$. Letting $n \to \infty$, either $\mathbb{P}(T < \infty) < 1$ and $\mathbb{E}_\infty[T] = \infty$, or $\mathbb{E}_\infty[T] \ge 1/\alpha$.

(b) Delay.. Condition on $\mathcal{F}_\tau$. For every j, $R_{\tau+n}^{(j)} \ge \prod_{k=\tau+1}^{\tau+n} e_k^{(j)}$ because the recursion contains the term started at $\tau+1$ with weight one, so $R_{\tau+n} \ge \nu_j \prod_{k=\tau+1}^{\tau+n} e_k^{(j)}$ with $\nu_j = 1/m$.

Take $\lambda \le \lambda^\star = \min\{\Delta/(2v), 1/(2w)\}$ on the grid, so $\lambda|X_k| \le \frac{1}{2}$ and $\log(1+x) \ge x - x^2$ for $x \ge -\frac{1}{2}$ gives $\log e_k \ge Y_k := \lambda X_k - \lambda^2 X_k^2$. Then $\mathbb{E}[Y_k \mid \mathcal{F}_{k-1}] \ge \lambda\Delta - \lambda^2 v \ge \lambda\Delta/2$ since $\lambda v \le \Delta/2$. Let $D_k = Y_k - \mathbb{E}[Y_k \mid \mathcal{F}_{k-1}]$. Because $|Y_k| \le \lambda w + \lambda^2 w^2 \le 1.5\lambda w$, $|D_k| \le 3\lambda w$. Because $Y_k^2 \le 2\lambda^2 X_k^2(1 + \lambda^2 X_k^2) \le 2.5\lambda^2 X_k^2$, $\mathrm{Var}(D_k \mid \mathcal{F}_{k-1}) \le 2.5\lambda^2 v$. Freedman's inequality (Freedman, 1975) for the lower tail with $s = n\lambda\Delta/4$ gives

$$\mathbb{P}\Big(\sum_{k=\tau+1}^{\tau+n} D_k \le -s \;\Big|\; \mathcal{F}_\tau\Big) \le \exp\Big(-\frac{s^2}{2(2.5n\lambda^2 v + 3\lambda w s/3)}\Big) = \exp\Big(-\frac{n\Delta^2}{80v + 8w\Delta}\Big).$$

This is at most δ when $n \ge (80v + 8w\Delta)\log(1/\delta)/\Delta^2$. On the complement, $\sum_k Y_k \ge n\lambda\Delta/2 - n\lambda\Delta/4 = n\lambda\Delta/4$.

The linear grid contains $\lambda_j = j/(mw) \in (\lambda^\star/2, \lambda^\star]$ with $j = \lfloor \lambda^\star mw \rfloor \ge 1$, which holds because $\lambda^\star \ge 1/(mw)$ under $\Delta \ge 2v/(mw)$ and $m \ge 2$. With $\lambda = \lambda_j$, on the good event $\log R_{\tau+n} \ge n\lambda_j\Delta/4 - \log m \ge n\lambda^\star\Delta/8 - \log m$, which is at least $\log(1/\alpha)$ when $n \ge 8\log(m/\alpha)/(\lambda^\star\Delta)$. Hence $T \le \tau + n$.

(c). Horizon h updates once per h rounds on a chain; apply (b) to the chain with its margin and second moment, and add at most h rounds for the position of the changepoint within a chain step. □

Appendix B. Experimental details

Simulator.. Linear-Gaussian latent system $z_{t+1} = Az_t + Ba_t + w_t$, $w_t \sim \mathcal{N}(0, 0.1\, I_4)$, spectral radius of A equal to 0.9, exogenous exploration actions $a_t \sim \mathcal{N}(0, 0.3^2 I)$, fixed unit drift direction u. The model uses the pre-change parameters, so it is exactly faithful until an unmodelled force fu is added to the true update. The statistic is $Z = \frac{1}{d}(y-\mu)^\top \Sigma_h^{-1}(y-\mu) - 1$ clipped to $[-1, 6]$ with the exact h-step predictive covariance $\Sigma_h = \sum_{i<h} A^i Q (A^i)^\top$.

Closed-loop simulator.. Same dynamics with state-dependent noise $q(z) = q_0(1 + \beta\|z\|^2/(\|z\|^2 + d))$, $q_0 = 0.1$, $\beta = 4$, a fully actuated $B = 0.8\, Q_{\mathrm{orth}}$, and certainty-equivalent one-step model-predictive control through the model (`atlas/closedloop.py`); actions are clipped to ± 2 with exploration noise 0.05.

Bets and thresholds.. Mixture over $m = 20$ equally spaced λ on $(0, 1/(z_{\mathrm{hi}} - z_{\mathrm{lo}})]$; level $\alpha = 0.05$, per-horizon level α/H.

Reproduction.. All simulator experiments run on CPU in minutes (`experiments/`, results in `results/`); the real-video replication needs the public Moving MNIST and KTH files.

Control study.. State (x, y, v_x, v_y), force $a \in [-1, 1]^2$, step 0.1: $p' = p + 0.1v$, $v' = v + 0.1(Ga - 0.6v - 0.5\|v\|v) + \sqrt{0.1}\,0.15\,\epsilon$. The target is $(\cos \omega t, \sin \omega t)$ with period 80 steps; stage cost $\|p - p^\star\|^2 + 0.01\|a\|^2$. The world model is an MLP ($6 \to 64 \to 64 \to 4$, tanh) predicting the next velocity increment and log-variance, trained for 40 epochs on 20,000 random-action transitions. The planner minimizes the model's mean-dynamics cost over the committed block with 25 Adam steps (terminal tracking weighted $3\times$). ATLAS statistics use 200 sampled rollouts per chain element and the sample covariance on the subspace with eigenvalues above 10^{-5}; statistics are clipped to $[-1, 8]$, tolerances are $\bar{z}_h + k\, s_h$ from 12 nominal episodes per model, and the frontier runs at level $\alpha/5$ per horizon. The CUSUM gate monitors the one-step standardized residual (reference mean and slack 0.25 sd from the same episodes; threshold chosen as the smallest on a grid with at most 5% false alarms over the calibration episodes). Faults begin at step 200 of 400.

Real-video models.. PCA-plus-linear: 20 (Moving MNIST) or 32 (KTH) components with a ridge-regularized linear transition. Convolutional autoencoder: three strided conv layers to a 20/32-dimensional code (6 epochs), then a residual MLP dynamics network (8 epochs); per-horizon covariance re-estimated on calibration clips with shrinkage 0.2. Frozen-encoder model: ImageNet ResNet-18 up to `layer3` with global pooling, PCA to 32 dimensions, residual MLP dynamics (25 epochs), same recalibration.

CRediT authorship contribution statement

Quang-Vinh Dang: Conceptualization, Methodology, Formal analysis, Software, Investigation, Validation, Visualization, Writing – original draft, Writing – review and editing.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data and code availability

All experiments are reproducible from the open-source implementation at `https://github.com/vinhqdang/ATLAS-Anytime-valid-Testing-of-LAtent-Simulators`, together with the public Moving MNIST and KTH datasets. Simulator and control experiments generate their own data; per-experiment outputs are in the repository.

References

Adams, R.P., MacKay, D.J.C., 2007. Bayesian online changepoint detection. arXiv preprint arXiv:0710.3742 doi:10.48550/arXiv.0710.3742.

Arnold, S., Henzi, A., Ziegel, J.F., 2023. Sequentially valid tests for forecast calibration. The Annals of Applied Statistics 17, 1909–1935. doi:10.1214/22-AOAS1697.

Assran, M., Bardes, A., Fan, D., Garrido, Q., Howes, R., Komeili, M., Muckley, M., Rizvi, A., Roberts, C., Sinha, K., Zholus, A., Arnaud, S., Gejji, A., Martin, A., Hogan, F.R., Dugas, D., Bojanowski, P., Khalidov, V., Labatut, P., Massa, F., Szafraniec, M., Krishnakumar, K., Li, Y., Ma, X., Chandar, S., Meier, F., LeCun, Y., Rabbat, M., Ballas, N., 2025. V-JEPA 2: Self-supervised video models enable understanding, prediction and planning. arXiv preprint arXiv:2506.09985 doi:10.48550/arXiv.2506.09985.

Assran, M., Duval, Q., Misra, I., Bojanowski, P., Vincent, P., Rabbat, M., LeCun, Y., Ballas, N., 2023. Self-supervised learning from images with a joint-embedding predictive architecture (i-jepa), in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). doi:10.48550/arXiv.2301.08243.

Bardes, A., Garrido, Q., Ponce, J., Chen, X., Rabbat, M., LeCun, Y., Assran, M., Ballas, N., 2024. Revisiting feature prediction for learning visual representations from video (v-jepa). arXiv preprint arXiv:2404.08471 doi:10.48550/arXiv.2404.08471.

Bifet, A., Gavaldà, R., 2007. Learning from time-changing data with adaptive windowing, in: SIAM International Conference on Data Mining (SDM). doi:10.1137/1.9781611972771.42.

Bruce, J., Dennis, M., Edwards, A., Parker-Holder, J., Shi, Y., Hughes, E., Lai, M., Mavalankar, A., Steigerwald, R., Apps, C., Aytar, Y., Bechtle, S., Behbahani, F., Chan, S., Heess, N., Gonzalez, L., Osindero, S., Ozair, S., Reed, S., Zhang, J., Zolna, K., Clune, J., de Freitas, N., Singh, S., Rocktäschel, T., 2024. Genie: Generative interactive environments, in: International Conference on Machine Learning (ICML). doi:10.48550/arXiv.2402.15391.

Choe, Y.J., Ramdas, A., 2024. Comparing sequential forecasters. Operations Research 72, 1368–1387. doi:10.1287/opre.2021.0792.

Chugg, B., Cortes-Gomez, S., Wilder, B., Ramdas, A., 2023. Auditing fairness by betting, in: Advances in Neural Information Processing Systems.

Fan, X., Grama, I., Liu, Q., 2015. Exponential inequalities for martingales and applications. Electronic Journal of Probability 20, 1–22.

Freedman, D.A., 1975. On tail probabilities for martingales. The Annals of Probability 3, 100–118.

Gibbs, I., Candès, E., 2021. Adaptive conformal inference under distribution shift, in: Advances in Neural Information Processing Systems (NeurIPS). doi:10.48550/arXiv.2106.00170.

Gibbs, I., Candès, E., 2024. Conformal inference for online prediction with arbitrary distribution shifts. Journal of Machine Learning Research doi:10.48550/arXiv.2208.08401.

Gneiting, T., Raftery, A.E., 2007. Strictly proper scoring rules, prediction, and estimation. Journal of the American Statistical Association 102, 359–378. doi:10.1198/016214506000001437.

Grünwald, P., de Heide, R., Koolen, W.M., 2024. Safe testing. Journal of the Royal Statistical Society: Series B 86, 1091–1128. doi:10.1093/jrsssb/qkae011.

Ha, D., Schmidhuber, J., 2018. World models. arXiv preprint arXiv:1803.10122 doi:10.48550/arXiv.1803.10122.

Hafner, D., Pasukonis, J., Ba, J., Lillicrap, T., 2023. Mastering diverse domains through world models. arXiv preprint arXiv:2301.04104 doi:10.48550/arXiv.2301.04104.

Henzi, A., Ziegel, J.F., 2022a. Correction to: Valid sequential inference on probability forecast performance. Biometrika 109, 1181–1182. doi:10.1093/biomet/asac043.

Henzi, A., Ziegel, J.F., 2022b. Valid sequential inference on probability forecast performance. Biometrika 109, 647–663. doi:10.1093/biomet/asab047.

Howard, S.R., Ramdas, A., McAuliffe, J., Sekhon, J., 2021. Time-uniform, nonparametric, nonasymptotic confidence sequences. The Annals of Statistics 49, 1055–1080. doi:10.1214/20-AOS1991.

Janner, M., Fu, J., Zhang, M., Levine, S., 2019. When to trust your model: Model-based policy optimization, in: Advances in Neural Information Processing Systems (NeurIPS). doi:10.48550/arXiv.1906.08253.

Kearns, M., Singh, S., 2002. Near-optimal reinforcement learning in polynomial time. Machine Learning 49, 209–232. doi:10.1023/A:1017984413808.

Lai, T.L., 1998. Information bounds and quick detection of parameter changes in stochastic systems. IEEE Transactions on Information Theory 44, 2917–2929. doi:10.1109/18.737522.

Lobel, S., Gottesman, O., Konidaris, G., 2024. An optimal tightness bound for the simulation lemma, in: Reinforcement Learning Conference (RLC). doi:10.48550/arXiv.2406.16249.

Lorden, G., 1971. Procedures for reacting to a change in distribution. The Annals of Mathematical Statistics 42, 1897–1908. doi:10.1214/aoms/1177693055.

NVIDIA, Agarwal, N., Ali, A., Bala, M., Balaji, Y., et al., 2025. Cosmos world foundation model platform for physical AI. arXiv preprint arXiv:2501.03575 doi:10.48550/arXiv.2501.03575.

Page, E.S., 1954. Continuous inspection schemes. Biometrika 41, 100–115. doi:10.2307/2333009.

Podkopaev, A., Ramdas, A., 2022. Tracking the risk of a deployed model and detecting harmful distribution shifts, in: International Conference on Learning Representations.

Raab, C., Heusinger, M., Schleif, F.M., 2020. Reactive soft prototype computing for concept drift streams. Neurocomputing 416, 340–351. doi:10.1016/j.neucom.2019.11.111.

Ramdas, A., Grünwald, P., Vovk, V., Shafer, G., 2023. Game-theoretic statistics and safe anytime-valid inference. Statistical Science 38, 576–601. doi:10.1214/23-STS894.

Schuldt, C., Laptev, I., Caputo, B., 2004. Recognizing human actions: A local SVM approach, in: International Conference on Pattern Recognition (ICPR). doi:10.1109/ICPR.2004.1334462.

Shafer, G., 2021. Testing by betting: A strategy for statistical and scientific communication. Journal of the Royal Statistical Society: Series A 184, 407–431. doi:10.1111/rssa.12647.

Shafer, G., Vovk, V., 2019. Game-Theoretic Foundations for Probability and Finance. Wiley. doi:10.1002/9781118548035.

Shekhar, S., Ramdas, A., 2024. Nonparametric two-sample testing by betting. IEEE Transactions on Information Theory .

Shin, J., Ramdas, A., Rinaldo, A., 2024. E-detectors: A nonparametric framework for sequential change detection. The New England Journal of Statistics in Data Science doi:10.48550/arXiv.2203.03532.

Srivastava, N., Mansimov, E., Salakhutdinov, R., 2015. Unsupervised learning of video representations using LSTMs, in: International Conference on Machine Learning (ICML). doi:10.48550/arXiv.1502.04681.

Tibshirani, R.J., Barber, R.F., Candès, E., Ramdas, A., 2019. Conformal prediction under covariate shift. Advances in Neural Information Processing Systems (NeurIPS) doi:10.48550/arXiv.1904.06019.

Unterthiner, T., van Steenkiste, S., Kurach, K., Marinier, R., Michalski, M., Gelly, S., 2018. Towards accurate generative models of video: A new metric and challenges. arXiv preprint arXiv:1812.01717 doi:10.48550/arXiv.1812.01717.

Ville, J., 1939. Étude critique de la notion de collectif. Gauthier-Villars, Paris .

Vovk, V., Gammerman, A., Shafer, G., 2005. Algorithmic Learning in a Random World. Springer. doi:10.1007/b106715.

Vovk, V., Nouretdinov, I., Gammerman, A., 2003. Testing exchangeability on-line, in: International Conference on Machine Learning.

Vovk, V., Wang, R., 2021. E-values: Calibration, combination and applications. The Annals of Statistics 49, 1736–1754. doi:10.1214/20-AOS2020.

Waudby-Smith, I., Ramdas, A., 2024. Estimating means of bounded random variables by betting. Journal of the Royal Statistical Society: Series B 86, 1–27. doi:10.1093/jrsssb/qkad009.

Table 5: Replicated real-video study (30 random splits per shift). False alarms over L in-distribution clips and over $3L$; detection rate and delay in clips after the shift; bootstrap 95% intervals in brackets. k is the tolerance margin in standard deviations of the calibration statistic. Tunable detectors are tuned to 5% on separate repetitions.

	FA @ L	FA @ $3L$	detection rate	delay (clips)
Moving MNIST: native → double speed				
ATLAS, $k = 0$	0.17 [0.03,0.30]	0.43 [0.27,0.60]	1.00	4.3 [3.8,4.8]
ATLAS, $k = 0.25$	0.00 [0.00,0.00]	0.00 [0.00,0.00]	1.00	4.6 [4.1,5.1]
ATLAS, $k = 0.5$	0.00 [0.00,0.00]	0.00 [0.00,0.00]	1.00	5.2 [4.6,5.8]
ATLAS, $k = 1$	0.00 [0.00,0.00]	0.00 [0.00,0.00]	1.00	6.6 [5.8,7.5]
CUSUM	0.07 [0.00,0.17]	0.43 [0.27,0.60]	1.00	3.2 [2.7,3.7]
ADWIN	0.07 [0.00,0.17]	0.37 [0.20,0.53]	0.00	—
KSWIN	0.03 [0.00,0.10]	0.27 [0.13,0.43]	0.00	—
Conformal martingale	0.53 [0.37,0.70]	0.67 [0.50,0.83]	1.00	0.3 [0.1,0.5]
KTH: walking → running				
ATLAS, $k = 0$	0.43 [0.27,0.60]	0.47 [0.30,0.63]	1.00	0.3 [0.1,0.5]
ATLAS, $k = 0.25$	0.17 [0.03,0.30]	0.37 [0.20,0.53]	1.00	0.5 [0.2,0.9]
ATLAS, $k = 0.5$	0.07 [0.00,0.17]	0.13 [0.03,0.27]	1.00	1.3 [0.5,2.2]
ATLAS, $k = 1$	0.03 [0.00,0.10]	0.03 [0.00,0.10]	0.70	5.3 [1.7,9.7]
CUSUM	0.97 [0.90,1.00]	0.93 [0.83,1.00]	1.00	0.1 [0.0,0.3]
ADWIN	0.17 [0.03,0.30]	0.43 [0.27,0.60]	0.00	—
KSWIN	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00	—
Conformal martingale	0.70 [0.53,0.87]	0.63 [0.47,0.80]	1.00	0.9 [0.5,1.4]
KTH: outdoor → indoor walking				
ATLAS, $k = 0$	0.33 [0.17,0.50]	0.50 [0.33,0.67]	1.00	1.9 [1.0,3.1]
ATLAS, $k = 0.25$	0.07 [0.00,0.17]	0.23 [0.10,0.40]	0.93	3.1 [1.6,5.2]
ATLAS, $k = 0.5$	0.07 [0.00,0.17]	0.03 [0.00,0.10]	0.77	4.1 [1.7,6.8]
ATLAS, $k = 1$	0.03 [0.00,0.10]	0.03 [0.00,0.10]	0.33	2.3 [0.6,4.5]
CUSUM	0.57 [0.40,0.73]	0.63 [0.47,0.80]	1.00	1.4 [0.9,2.0]
ADWIN	0.10 [0.00,0.20]	0.37 [0.20,0.53]	1.00	6.2 [5.2,7.2]
KSWIN	0.00 [0.00,0.00]	0.00 [0.00,0.00]	1.00	5.6 [4.2,7.2]
Conformal martingale	0.43 [0.27,0.60]	0.57 [0.40,0.73]	1.00	0.4 [0.2,0.6]

Table 6: Three world-model classes on the same real shifts. Each cell: false-alarm rate over $3L$ in-distribution clips / detection rate / mean delay in clips ("–" if never detected); repetitions in parentheses. ADWIN and KSWIN, tuned to 5% on separate repetitions, almost never detected a shift with the neural models (one exception: ADWIN in 2 of 10 repetitions of the conv. autoencoder on the domain shift) and are omitted.

shift	world model	ATLAS k=0.5	ATLAS k=1	CUSUM	conformal MG
Moving MNIST speed	PCA + linear (30 splits)	0.00 / 1.00 / 5.2	0.00 / 1.00 / 6.6	0.43 / 1.00 / 3.2	0.67 / 1.00 / 0.3
	conv. autoencoder (10)	0.00 / 0.70 / 51.0	0.00 / 0.20 / 8.0	0.50 / 1.00 / 17.8	1.00 / 1.00 / 0.7
KTH walking→running	PCA + linear (30)	0.13 / 1.00 / 1.3	0.03 / 0.70 / 5.3	0.93 / 1.00 / 0.1	0.63 / 1.00 / 0.9
	conv. autoencoder (10)	0.40 / 0.50 / 10.0	0.40 / 0.40 / 12.5	0.50 / 0.60 / 8.8	0.90 / 0.80 / 0.4
	frozen ResNet + dynamics (10)	0.30 / 1.00 / 1.5	0.00 / 1.00 / 3.0	0.60 / 1.00 / 1.0	0.70 / 1.00 / 0.2
KTH outdoor→indoor	PCA + linear (30)	0.03 / 0.77 / 4.1	0.03 / 0.33 / 2.3	0.63 / 1.00 / 1.4	0.57 / 1.00 / 0.4
	conv. autoencoder (10)	0.10 / 0.60 / 15.5	0.10 / 0.30 / 12.3	0.20 / 0.30 / 11.3	0.60 / 0.50 / 0.0
	frozen ResNet + dynamics (10)	0.30 / 0.90 / 5.7	0.00 / 0.90 / 8.3	0.30 / 0.40 / 13.5	0.50 / 0.70 / 8.9

Table 7: Simulator, $\alpha = 0.05$: false-alarm rate at the calibration horizon $T_{\text{cal}} = 400$ and at $3T_{\text{cal}}$, and mean detection delay (rounds). BOCPD tuned to the false-alarm target does not detect this drift.

	monitor	FA @ T_{cal}	FA @ $3T_{\text{cal}}$	delay
anytime-valid	**ATLAS**	**0.000**	**0.003**	53.8
	Conformal martingale	0.023	0.023	77.9
tuned	CUSUM	0.113	0.273	12.2
	Page–Hinkley	0.050	0.527	30.4
	ADWIN	0.040	0.140	23.8
	KSWIN	0.040	0.087	33.4
	BOCPD	0.083	0.083	—
	e-detector	0.013	0.197	32.8

Table 8: Ablations. FA = false-revocation rate under the null (or, in B, revocation rate for a model whose true gap is $g = 0.0138$); detect = fraction detected after the drift; delay in rounds.

	variant	FA	detect	delay
A. bet	mixture (20 values)	.025	1.00	392
	plug-in (GRAPA)	.015	1.00	429
	fixed $\lambda = 0.1\lambda_{\max}$	.000	.97	1397
	fixed $\lambda = 0.5\lambda_{\max}$	.030	1.00	401
	fixed $\lambda = \lambda_{\max}$	.038	.99	362
B. tolerance	$\varepsilon = 0$	.125	—	—
	$\varepsilon = 0.5g$	.055	—	—
	$\varepsilon = g$	.025	—	—
	$\varepsilon = 1.5g$	.020	—	—
	$\varepsilon = 2.5g$	.005	—	—
C. drift shape	step	.030	.99	365
	ramp over 300 rounds	.030	.99	607
	transient pulse (150 rounds)	.030	.17	326
D. statistic	standardized excess	.025	1.00	392
	log-score vs. climatology, $\varepsilon = -1$	.000	.00	—
	log-score vs. climatology, $\varepsilon =$ pooled null mean	**.253**	.00	—
E. noise tails	Gaussian	.025	1.00	392
	Student-t_5 (variance-matched)	.007	.98	821